



UNIVERSITY OF WESTERN ATTICA

DEPARTMENT OF INFORMATION AND
COMPUTER ENGINEERING

DIGITAL DESIGN

WORKSHOP 2

ADDERS-DEDUCTORS

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CHAPTER 1

A FEW WORDS ABOUT THE WORK

The work is based on adders and subtractors. More specifically, the semi-summator, semi-factor, full adder and full subtractor. For each of them separately, the truth tables, the logical equations derived from them, the simulation part from the " multisim " simulator separately for different input values are presented and finally their implementation is described.

CHAPTER 2

BIBLIOGRAPHY

- Design and implementation of logic circuits P
. Giannakopoulos L. Aslanoglou

CHAPTER 3

DESCRIPTION OF THE IMPLEMENTATION OF THE WORK

For the implementation of the project the simulator " multisim " was used for the design of the circuits. In more detail, cables, grounds, VCC sources, lamps, logic gates (AND , OR , NOR , NAND , XOR , XNOR , NOT), oscilloscope were used.

CHAPTER 4

EXERCISES

EXERCISE 4.2.1 SEMI-AGGREGATE

THEORETICAL RESULTS

The truth matrix of the semi-integrator is:

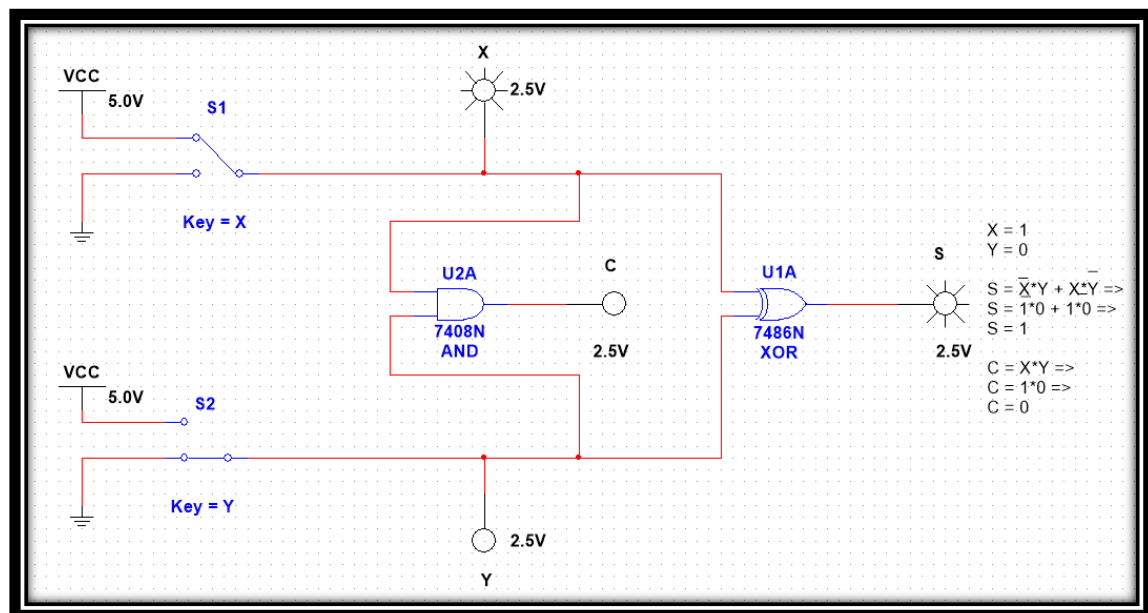
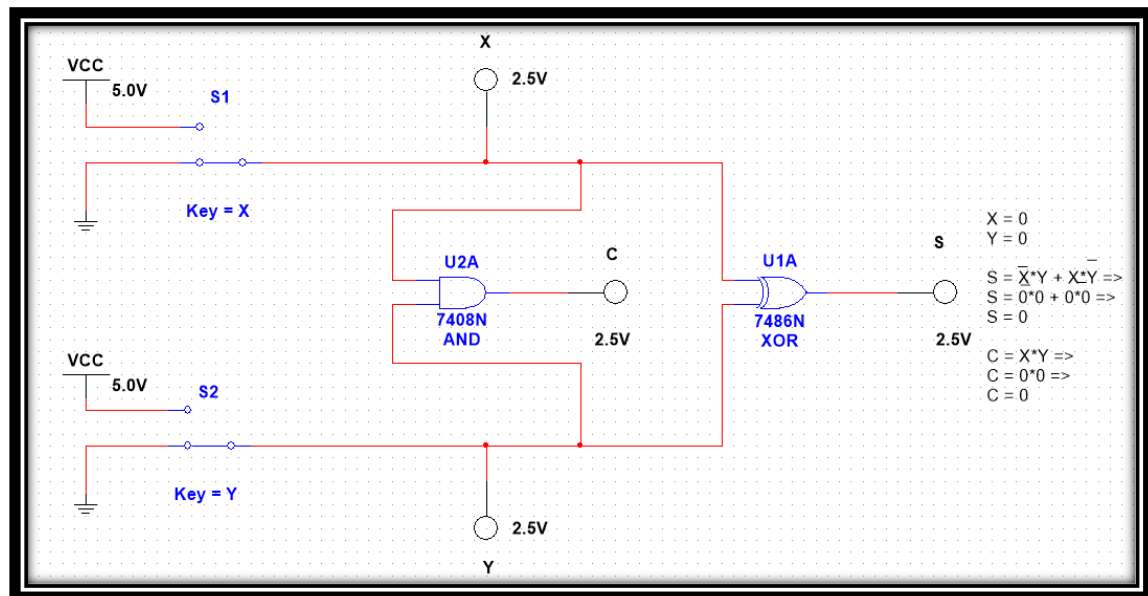
X	Y	S	C
0	0	0	0
1	0	1	0
0	1	1	0
1	1	0	1

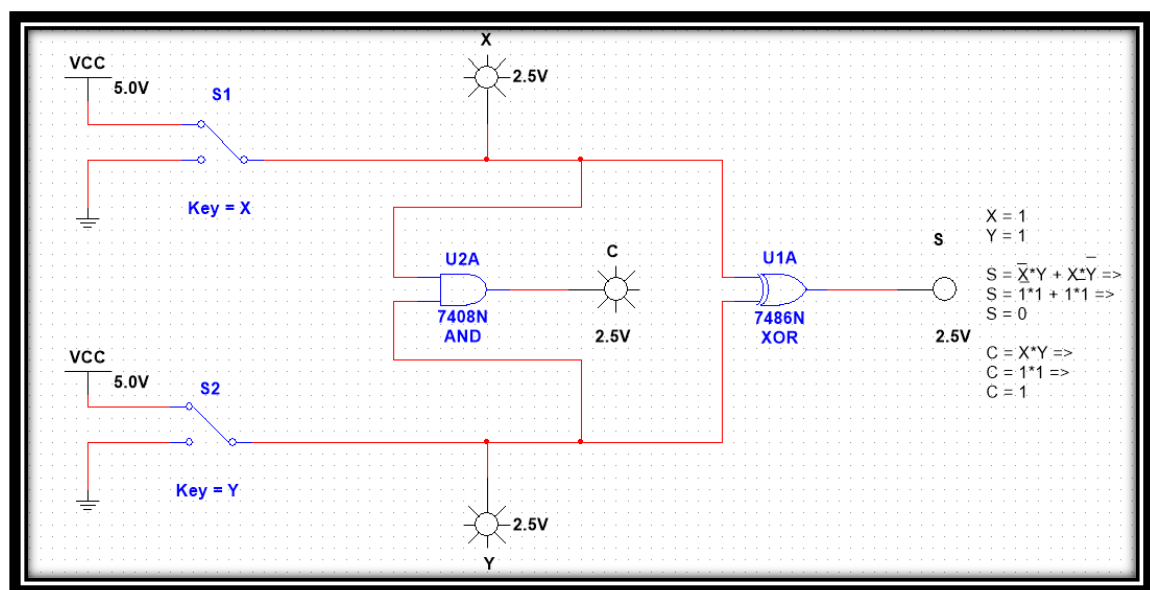
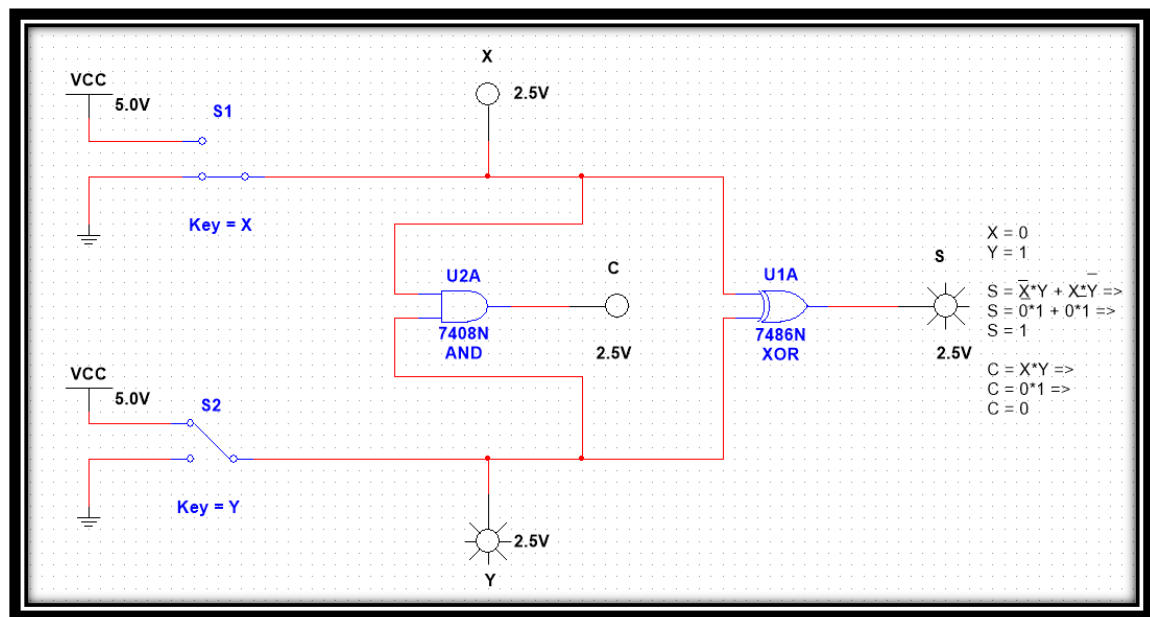
Where " X " and " Y " are two one-digit binary numbers to be summed and " S ", " C " are their sum and the possible holding respectively. The following logical equations are derived from the truth table:

$$S = \bar{X} * Y + X * \bar{Y}$$

$$C = X * Y$$

ADVANCE NOTIFICATION





IMPLEMENTATION

The implementation of the semi-integrator is done with one gate " AND " and one gate " XOR " and with two inputs "X", "Y". The voltages of the grounds and the VCC sources correspond to logic states. In general, a source with a voltage of 0 to 0.5 Volt corresponds to the logical state "0", while a source with a voltage of 2.7 to 5 Volt corresponds to the logical state "1". More specifically, grounding with 0 Volt corresponds to "0" and VCC source with 5 Volt corresponds to "1". So when the switch (S1, S2) is connected to the VCC source then the logic gate connected to the circuit gets an input value of logic "1" and a lamp (X,Y) of 2.5 Volt voltage which is also connected to the circuit lights up, whereas if the switch is connected to the ground then it gets an input value of "0" and the lamp does not light up. After the output of the logic gate, another lamp (S , C) also of 2.5 Volt voltage is connected, which depending on the type of logic gate and the potential connected to the two inputs, lights up or does not light up.

According to the above information and specifically for the logic gate " AND " that we use to calculate the possible holding of the two one-digit binary numbers we have:

- i) Lamp X : off / Lamp Y : off => => Lamp C : closed
- (ii) Lamp X : open / Lamp Y : closed => => Lamp C: closed Lamp C : closed
- iii) Lamp X : closed / Lamp Y : open => => Lamp C: open Lamp C : closed
- (iv) Lamp X : open / Lamp Y : open => Lamp C : open

According to the above information and specifically for the logic gate " XOR " that we use to calculate the sum of two single digit binary numbers we have:

- i) Lamp X : closed / Lamp Y : closed => => Lamp S : closed
- (ii) Lamp X : open / Lamp Y : closed => => Lamp S: closed Lamp S : open
- (iii) Lamp X : closed / Lamp Y : open => => Lamp S: open Lamp S : open
- (iv) Lamp X : open / Lamp Y : open => S Lamp S : closed

EXERCISE 4.2.2.2 SEMI-SUBTRACTOR

THEORETICAL RESULTS

The truth matrix of the semifractor is:

X	Y	D	B
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

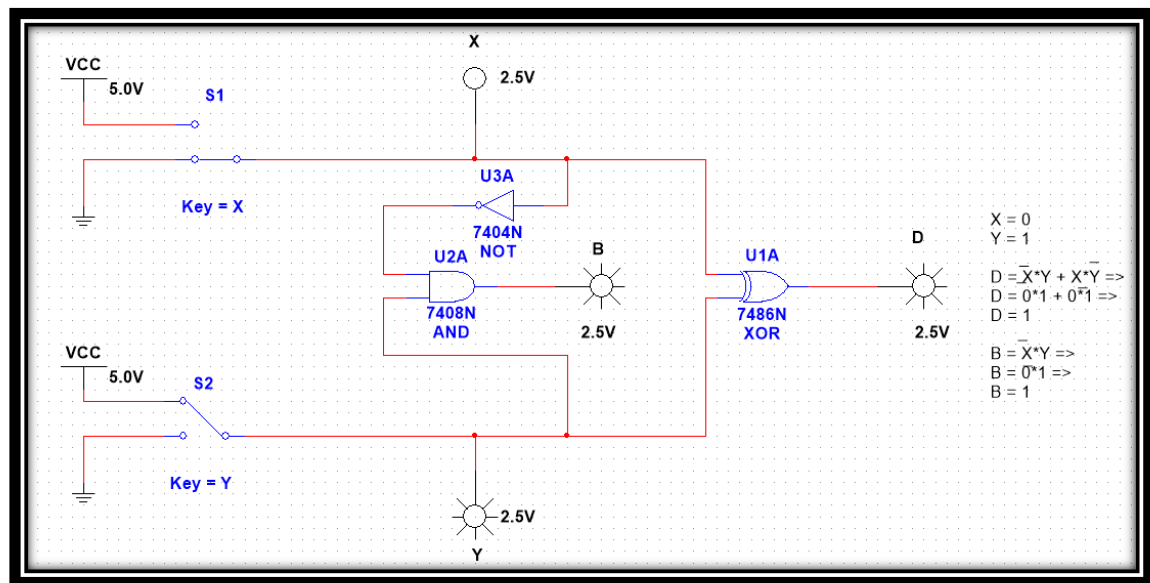
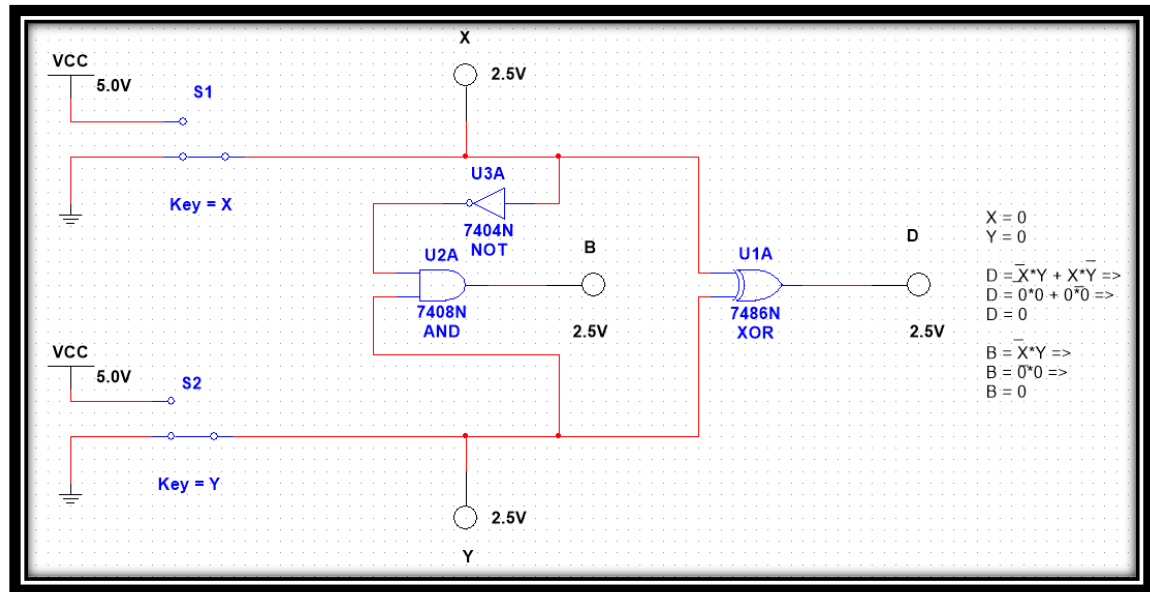
Where " X " and " Y " are two one-digit binary numbers to be subtracted and " D ", " B " their difference and the possible borrower respectively.

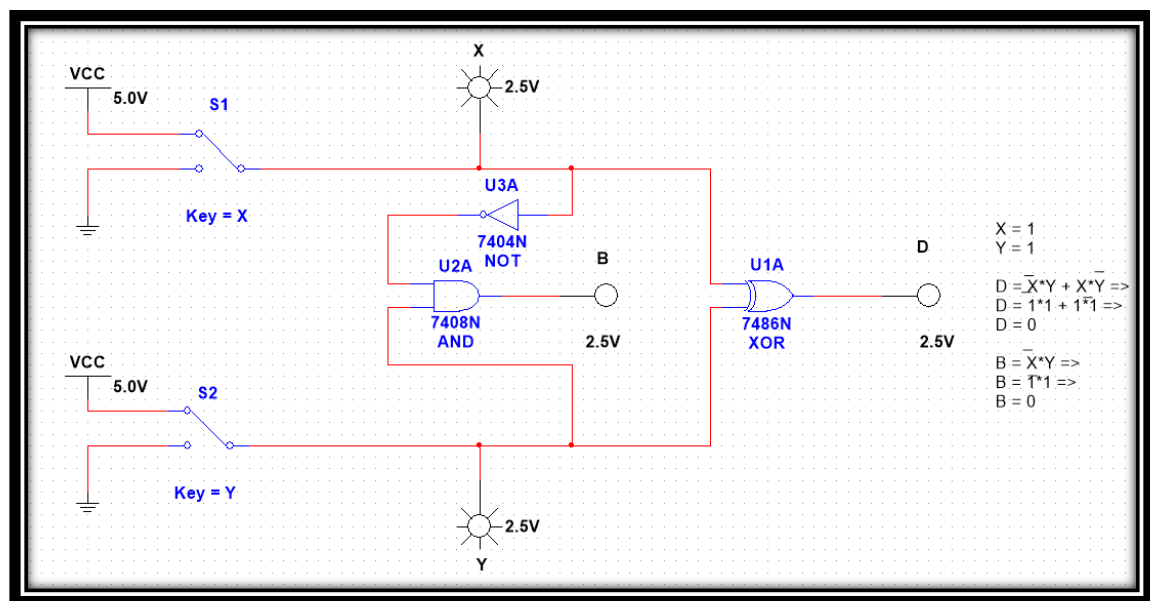
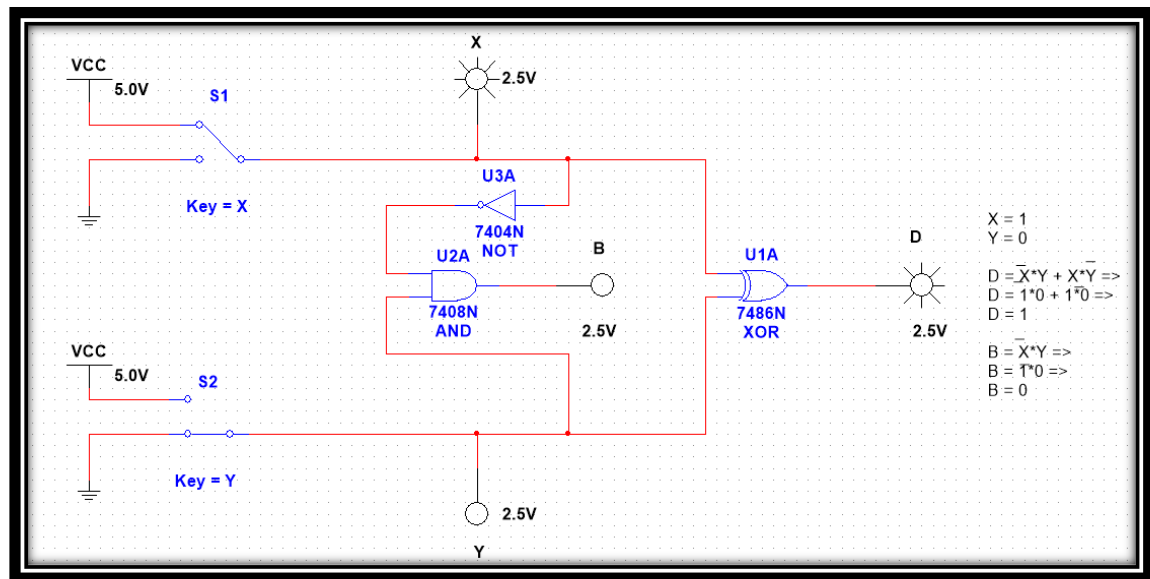
The following logical equations are derived from the truth table:

$$D = \bar{X} * Y + X * \bar{Y}$$

$$B = \bar{X} * Y$$

ADVANCE NOTIFICATION





IMPLEMENTATION

The implementation of the semi-divider is done with one gate " AND " one gate " NOT " at the "X" input of the gate " AND " and one gate " XOR " and two inputs " X ", "Y". The voltages of the VCC grounds and sources correspond to logic states. In general, a source with a voltage of 0 to 0.5 Volt corresponds to the logical state "0", while a source with a voltage of 2.7 to 5 Volt corresponds to the logical state "1". More specifically, grounding with 0 Volt corresponds to "0" and VCC source with 5 Volt corresponds to "1". So when the switch (S1, S2) is connected to the VCC source then the logic gate connected to the circuit gets an input value of logic "1" and a lamp (X,Y) of 2.5 Volt voltage which is also connected to the circuit lights up, whereas if the switch is connected to the ground then it gets an input value of "0" and the lamp does not light up. After the output of the logic gate, another lamp (D , B) also of 2.5 Volt voltage is connected, which depending on the type of logic gate and the potential connected to the two inputs, lights up or does not light up. According to the above information and specifically for the logic gate " AND ", in which there is a gate " NOT " at input "X" which we use to calculate the possible borrowing of the two one-digit binary numbers we have:

- i) Lamp X : closed / Lamp Y : closed $\Rightarrow \Rightarrow$ Lamp B: closed
 - (ii) Lamp X : open / Lamp Y : closed $\Rightarrow \Rightarrow$ Lamp X : open / Lamp Y : closed Lamp B: closed
 - iii) Lamp X : closed / Lamp Y : open $\Rightarrow \Rightarrow$ Lamp B: open Lamp B: open
 - (iv) Lamp X : open / Lamp Y : open $\Rightarrow \Rightarrow$ Lamp B: open Lamp B: closed
- According to the above information and specifically for the logic gate " XOR " that we use to calculate the sum of two single digit binary numbers we have:

- i) Lamp X : closed / Lamp Y : closed $\Rightarrow \Rightarrow$ Lamp D : closed
- (ii) Lamp X : open / Lamp Y : closed $\Rightarrow \Rightarrow$ Lamp D: closed Lamp D : open
- iii) Lamp X : closed / Lamp Y : open $\Rightarrow \Rightarrow$ Lamp D: open Lamp D : open
- (iv) Lamp X : open / Lamp Y : open $\Rightarrow$ Lamp D : closed

EXERCISE 4.2.3 FULL SUMMATION

THEORETICAL RESULTS

The truth table of the full adder is as follows:

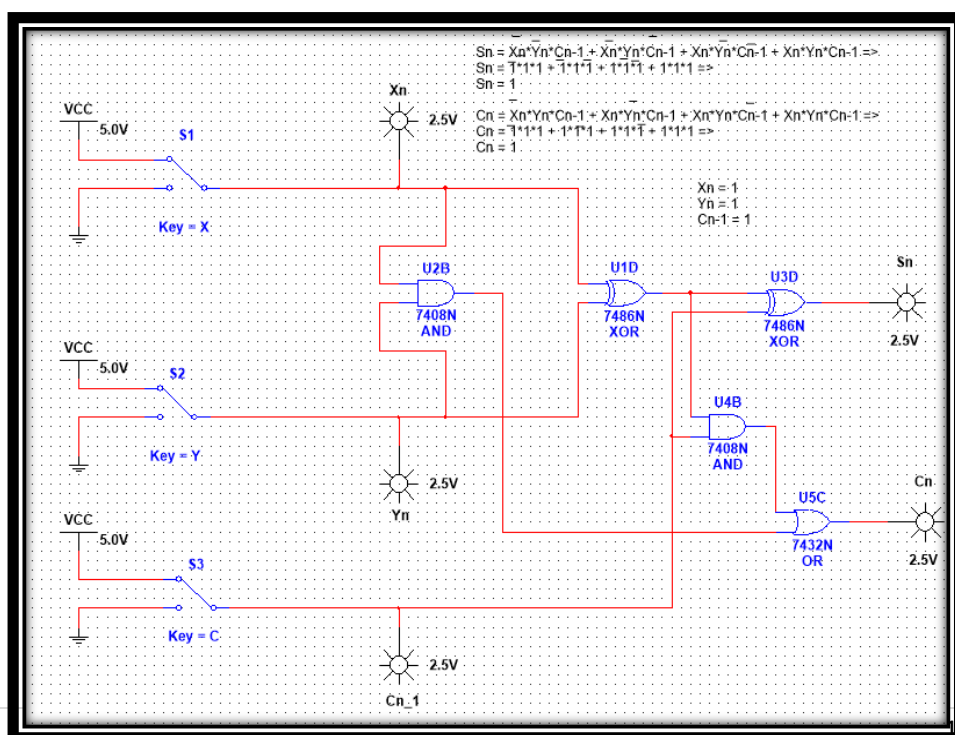
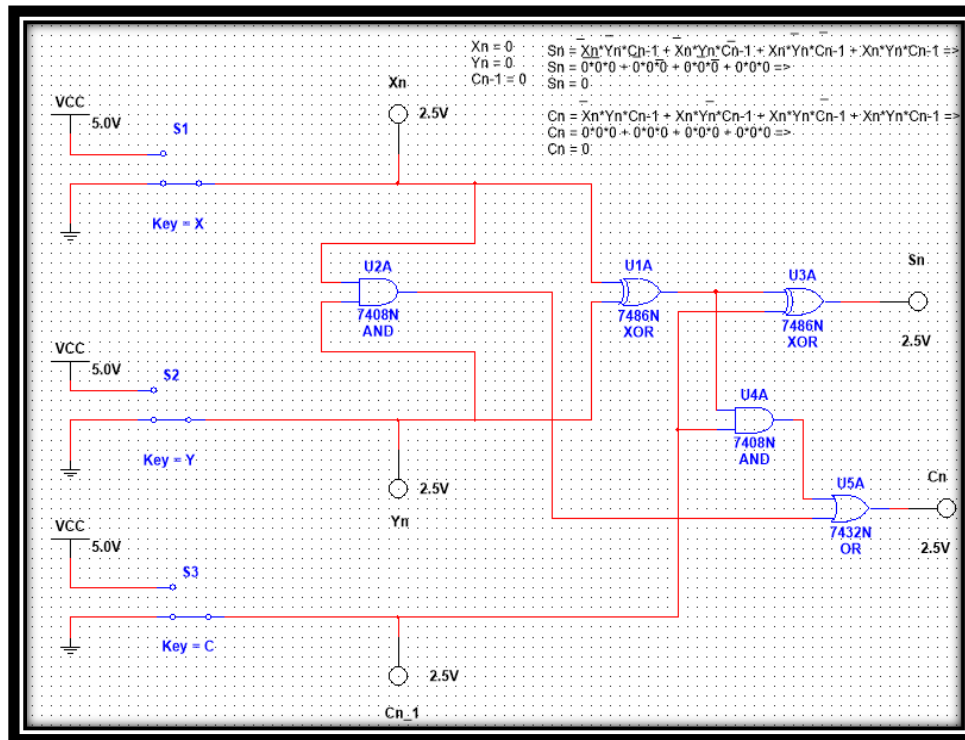
Xn	Yn	Cn-1	Sn	Cn
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

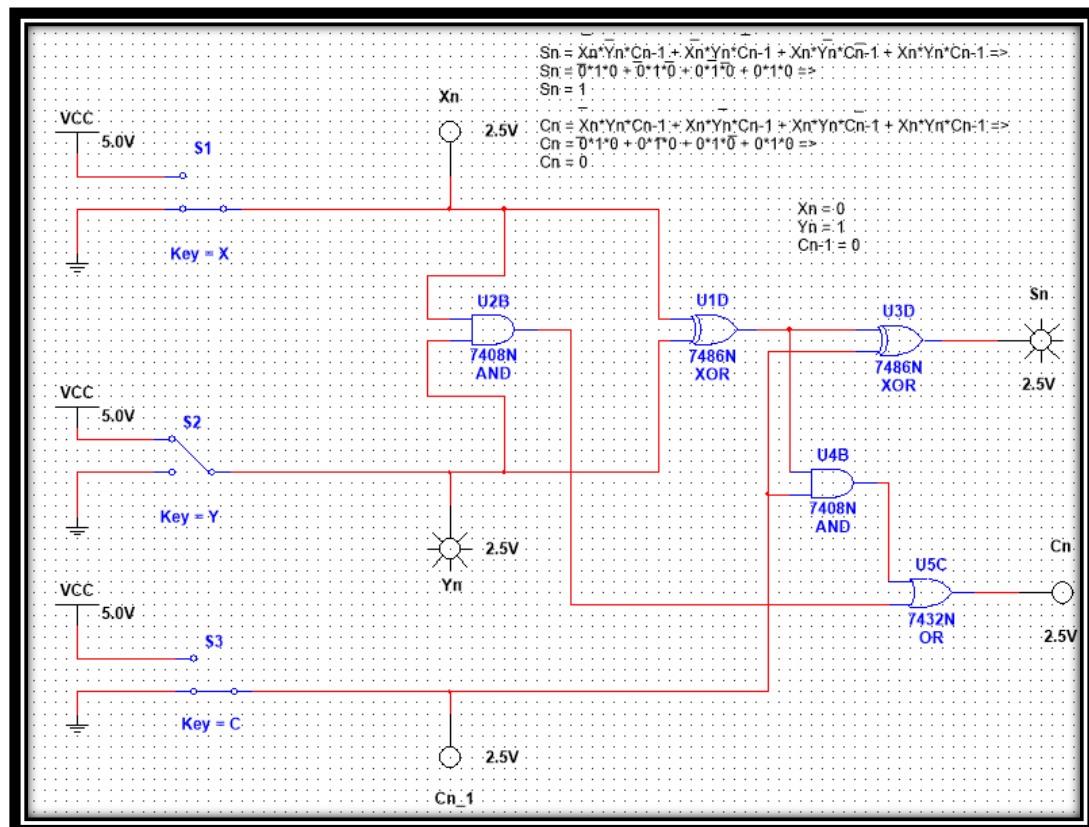
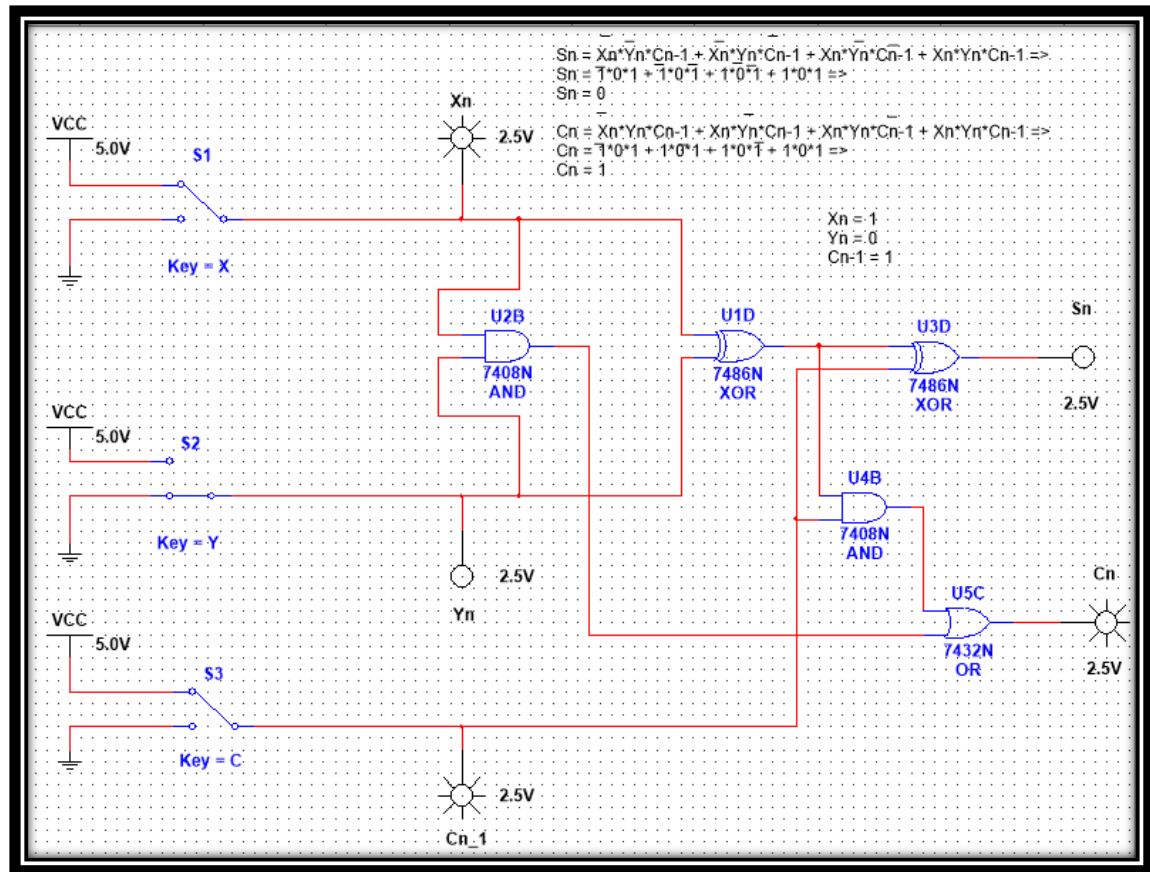
Where " Xn ", " Yn " are the digits in the " n " column and "Cn-1" is the possible hold from the summation of the digits in the previous column. We define " Sn " as the sum of " Xn ", " Yn " and "Cn-1" and " Cn " as the new possible holding. The logical equations resulting from the truth table are as follows:

$$S_n = \bar{X}_n \bar{Y}_n \bar{C}_{n-1} + \bar{X}_n \bar{Y}_n C_{n-1} + \bar{X}_n Y_n \bar{C}_{n-1} + X_n \bar{Y}_n \bar{C}_{n-1} + X_n \bar{Y}_n C_{n-1} + \bar{X}_n Y_n C_{n-1} + X_n Y_n \bar{C}_{n-1} + X_n Y_n C_{n-1}$$

$$C_n = \bar{X}_n \bar{Y}_n \bar{C}_{n-1} + \bar{X}_n \bar{Y}_n C_{n-1} + \bar{X}_n Y_n \bar{C}_{n-1} + X_n \bar{Y}_n \bar{C}_{n-1} + X_n \bar{Y}_n C_{n-1} + \bar{X}_n Y_n C_{n-1} + X_n Y_n \bar{C}_{n-1} + X_n Y_n C_{n-1}$$

ADVANCE NOTIFICATION





IMPLEMENTATION

The implementation of the full adder is done with two "AND" gates, two "XOR" gates, one "OR" gate and 3 inputs "Xn", "Yn" and "Cn-1". The voltages of the VCC grounds and sources correspond to logic states. In general, a source with a voltage of 0 to 0.5 Volt corresponds to the logical state "0", while a source with a voltage of 2.7 to 5 Volt corresponds to the logical state "1". More specifically, grounding with 0 Volt corresponds to "0" and VCC source with 5 Volt corresponds to "1". So when the switch (S1, S2, S3) is connected to the VCC source then the logic gate connected to the circuit gets an input value of logic "1" and a lamp (Xn, Yn, Cn-1) of 2.5 Volt voltage which is also connected to the circuit is lit, whereas if the switch is connected to the ground then it gets an input value of "0" and the lamp is not lit. After the output of the logic gate, another lamp (Sn, Cn) also of 2.5 Volt voltage is connected, which depending on the type of logic gate and the potential connected to the two inputs, lights up or does not light up.

According to the above information and specifically for the logic gate "AND U2B", which we use to calculate input 2 (bottom) of "OR U5C" we have:

- i) Lamp Xn : closed / Lamp Yn : closed => Input 2 OR U5C : 0
- ii) Lamp Xn : open / Lamp Yn : closed => Input 2 OR U5C : 0
- (iii) Lamp Xn : closed / Lamp Yn : open => Input 2 OR U5C : 0
- iv) Lamp Xn : open / Lamp Yn : open => Input 2 OR U5C : 1

According to the above information and specifically for the logic gate "XOR U1D" that we use to calculate input 1 (up) of "XOR U3D" we have:

- i) Lamp Xn : closed / Lamp Yn : closed => Input 1 XOR U3D : 0
- ii) Lamp Xn : open / Lamp Yn : closed => Input 1 XOR U3D : 1
- (iii) Lamp Xn : closed / Lamp Yn : open => Input 1 XOR U3D : 1
- (iv) Lamp Xn : open / Lamp Yn : open => Input 1 XOR U3D : 0

According to the above information and specifically for the logic gate "XOR U3D" that we use to calculate the sum "Sn" of "Xn", "Yn" and "Cn-1" we have:

i) Output XOR U1D : 0 / Lamp Cn-1: closed => Lamp Sn : closed
(ii) Output XOR U1D : 1 / Lamp Cn- 1: closed => => XOR U1D : 1 / Lamp Cn-1 : closed Lamp Sn : open

(iii) Output XOR U1D : 0 / Lamp Cn-1: open => Sn Lamp Sn : open

iv) Output XOR U1D : 1 / Lamp Cn-1: open => Sn Lamp Sn : closed

According to the above information and specifically for the logic gate " AND U4B " used to calculate the input 1 (up) of " OR U5C " we have:

i) Lamp Cn-1: closed / Output XOR U1D : 0 => Input 1 OR U5C : 0

ii) Lamp Cn-1: open / Output XOR U1D : 0 => Input 1 OR U5C : 0

(iii) Lamp Cn-1: closed / Output XOR U1D : 1 => Input 1 OR U5C : 0

iv) Lamp Cn-1: open / Output XOR U1D : 1 => Input 1 OR U5C : 1

According to the above information and specifically for the logic gate " OR U5C " that we use to calculate the new possible hold " Cn " we have:

i) Output AND U2B : 0 / Output AND U4B : 0 => Lamp Cn : closed

ii) Output AND U2B : 1 / Output AND U4B : 0 => Lamp Cn : open

(iii) Output AND U2B : 0 / Output AND U4B : 1 => Lamp Cn : open

(iv) Output AND U2B : 1 / Output AND U4B : 1 => Lamp Cn : open

EXERCISE 4.2.4 COMPLETE SUBTRACTION

THEORETICAL RESULTS

The truth table of the full subtractor is as follows:

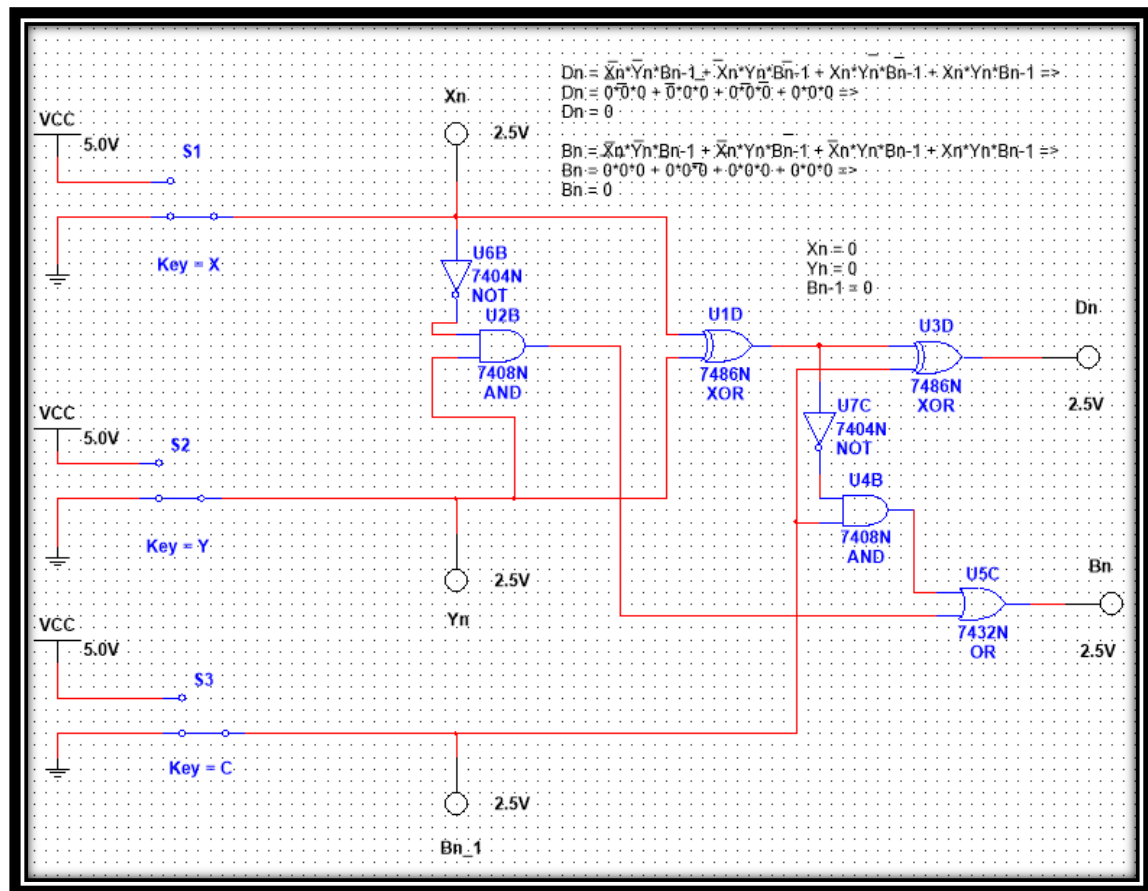
Xn	Yn	Bn-1	Dn	Bn
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

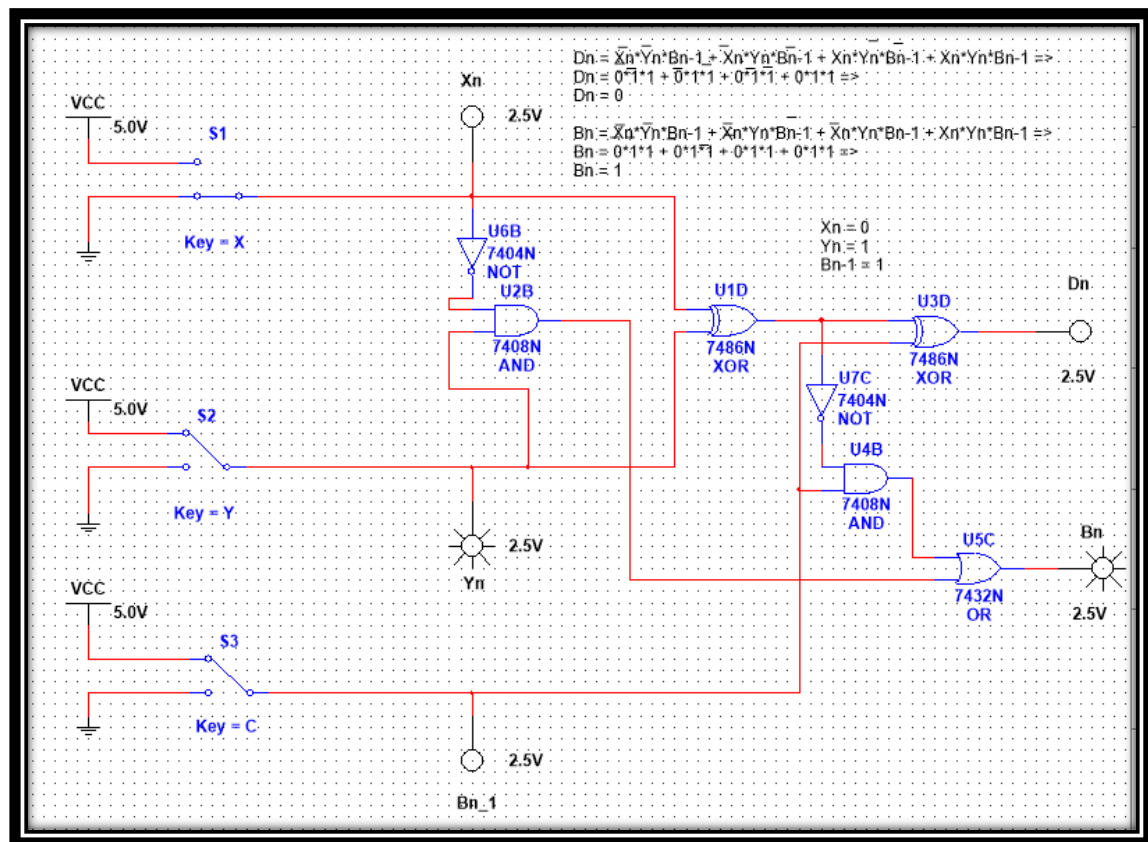
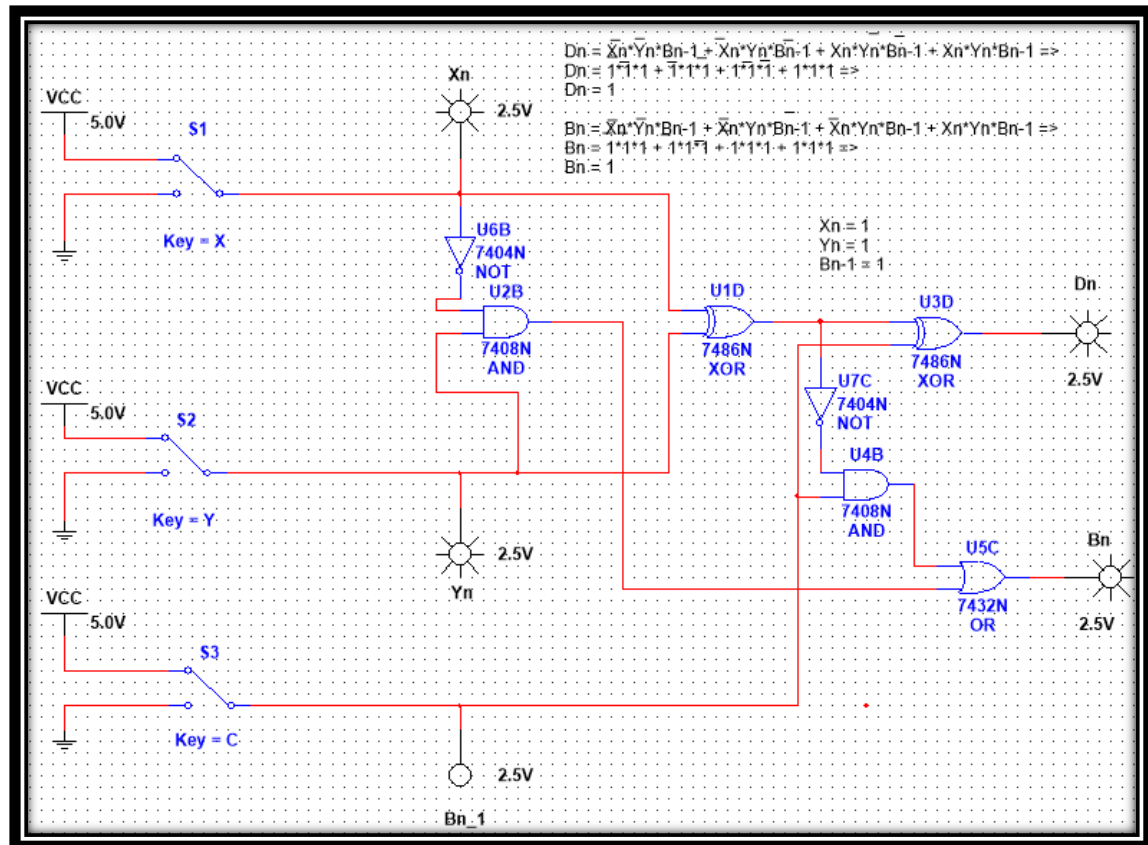
Where " Xn ", " Yn " are the digits the " n " digit of the decrement and the subtractor respectively and "Bn-1" is the possibly borrowed from the subtraction of the digits in the previous column. As " Sn " we define the difference between " Xn ", " Yn " and "Bn-1" and as " Bn " the new possible borrower. The logical equations resulting from the truth table are as follows:

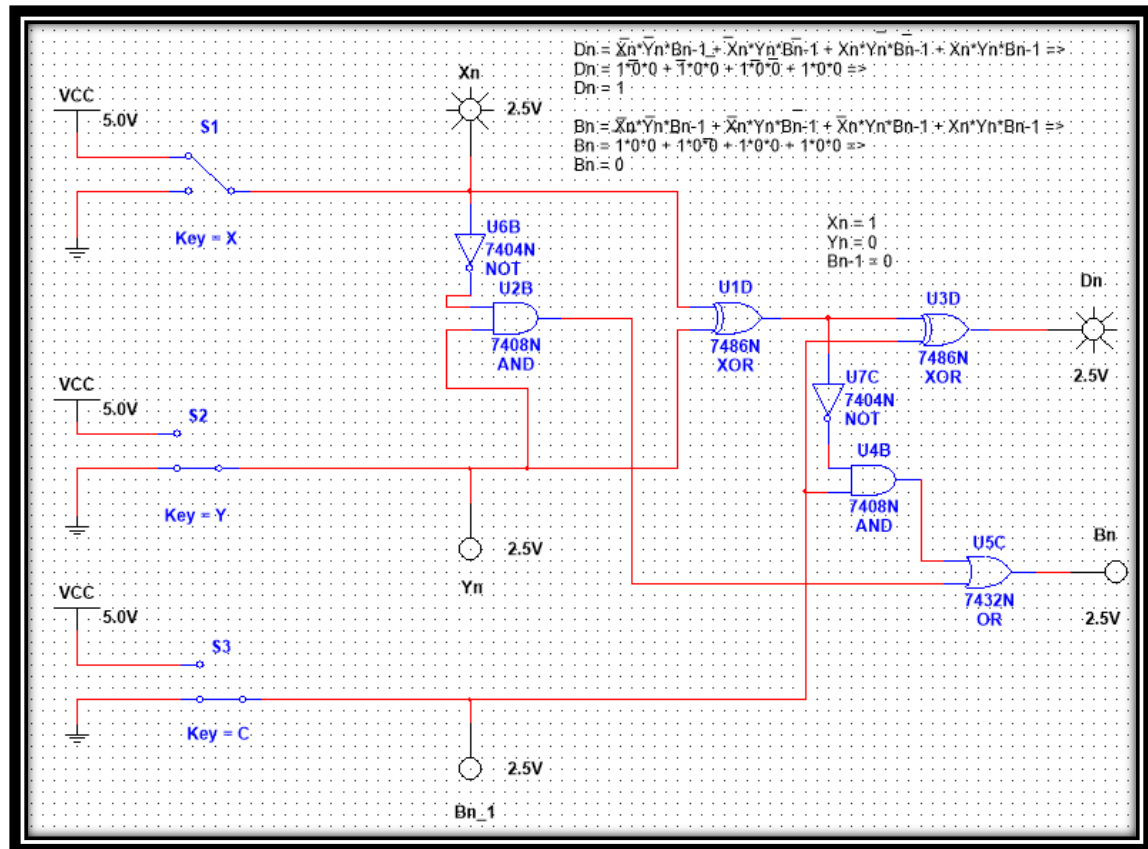
$$D_n = \bar{X}_n \bar{Y}_n B_{n-1} + \bar{X}_n Y_n \bar{B}_{n-1} + X_n \bar{Y}_n \bar{B}_{n-1} + X_n Y_n B_{n-1}$$

$$B_n = \bar{X}_n \bar{Y}_n B_{n-1} + \bar{X}_n Y_n B_{n-1} + X_n \bar{Y}_n B_{n-1} + X_n Y_n \bar{B}_{n-1}$$

ADVANCE NOTIFICATION







IMPLEMENTATION

The implementation of the complete abstractor is done with two " AND " gates to which two " NOT " gates are connected for each input, two " XOR " gates, one " OR " gate and 3 inputs " Xn ", " Yn " and " Bn-1 ". The voltages of the VCC grounds and sources correspond to logic states. In general, a source with a voltage of 0 to 0.5 Volt corresponds to the logic state "0", while a source with a voltage of 2.7 to 5 Volt corresponds to the logic state "1". More specifically, grounding with 0 Volt corresponds to "0" and VCC source with 5 Volt corresponds to "1". So when the switch (S1, S2, S3) is connected to the VCC source then the logic gate connected to the circuit gets an input value of logic "1" and a lamp (Xn , Yn , Bn-1) of 2.5 Volt voltage which is also connected to the circuit is lit, whereas if the switch is connected to the ground then it gets an input value of "0" and the lamp is not lit. After the output of the logic gate another lamp (Bn , Dn) also of 2.5 Volt voltage is connected, which

depending on the type of logic gate and the potential connected to the two inputs, is lit or not lit.

According to the above information and specifically for the logic gate " AND U2B ", where at input 1 (top) is connected a gate " NOT U6B ", which we use to calculate input 2 (bottom) of " OR U5C " we have:

- i) Lamp Xn : closed / Lamp Yn : closed $\Rightarrow$ Input 2 OR U5C : 0
- ii) Lamp Xn : open / Lamp Yn : closed $\Rightarrow$ Input 2 OR U5C : 0
- (iii) Lamp Xn : closed / Lamp Yn : open $\Rightarrow$ Input 2 OR U5C : 1
- (iv) Lamp Xn : open / Lamp Yn : open $\Rightarrow$ Input 2 OR U5C : 0

According to the above information and specifically for the logic gate " XOR U1D " that we use to calculate input 1 (up) of " XOR U3D " we have:

- i) Lamp Xn : closed / Lamp Yn : closed $\Rightarrow$ Input 1 XOR U3D : 0
- ii) Lamp Xn : open / Lamp Yn : closed $\Rightarrow$ Input 1 XOR U3D : 1
- (iii) Lamp Xn : closed / Lamp Yn : open $\Rightarrow$ Input 1 XOR U3D : 1
- (iv) Lamp Xn : open / Lamp Yn : open $\Rightarrow$ Input 1 XOR U3D : 0

According to the above information and specifically for the logic gate " XOR U3D " that we use to calculate the sum " Dn " of " Xn ", " Yn " and " Bn-1 " we have:

- i) Output XOR U1D : 0 / Lamp Bn-1: closed $\Rightarrow$ Lamp Dn : closed
- (ii) Output XOR U1D : 1 / Lamp Bn-1: closed $\Rightarrow$ XOR U1D : 1 / Lamp Bn-1 : closed Lamp Dn : open
- iii) Output XOR U1D : 0 / Lamp Bn- 1: open $\Rightarrow$ Dn-1 : open Lamp Dn : open
- (iv) Output XOR U1D : 1 / Lamp Bn-1: open $\Rightarrow$ Lamp Bn-1 : open Lamp Dn : closed

According to the above information and specifically for the logic gate " AND U4B ", where a gate " NOT U7C " is connected to input 1 (up), which we use to calculate input 1 (up) of " OR U5C " we have:

- i) Lamp Bn-1: closed / Output XOR U1D : 0 $\Rightarrow$ Input 1 OR U5C : 0
- ii) Lamp Bn-1: open / Output XOR U1D : 0 $\Rightarrow$ Input 1 OR U5C : 0
- (iii) Lamp Bn-1: closed / Output XOR U1D : 1 $\Rightarrow$ Input 1 OR U5C : 1
- (iv) Lamp Bn-1: open / Output XOR U1D : 1 $\Rightarrow$ Input 1 OR U5C : 0

According to the above information and specifically for the logic gate " OR U5C " that we use to calculate the new possible hold " Cn " we have:

- i) Output AND U2B : 0 / Output AND U4B : 0 $\Rightarrow$ Lamp Bn : closed

- ii) Output AND U2B : 1 / Output AND U4B : 0 => Lamp Bn : open
- (iii) Output AND U2B : 0 / Output AND U4B : 1 => Lamp Bn : open
- (iv) Output AND U2B : 1 / Output AND U4B : 1 => Lamp Bn : open